

(2A) Misconfiguration

(i) accept self-signed certificates

Location(s) in code:

client.py

```
context.load_verify_locations("server.crt")
```

Since we are self-signing our certificates and not using a third-party authority to truly authenticate our certificates, there is the possibility of malicious attackers providing fake certificates and from there being able to obtain sensitive information.

(ii) permit weak cipher suites

Location(s) in code:

Since there is no specific location in our code that explicitly sets a cipher, there are currently no misconfigurations in our code that can lead to weak cipher suites. After further research, it is known that python's SSL library will choose the most secure cipher suite based on both party compatibility.

However, if there were the possibility of configuring the use of a weak cipher (possibly under during the "conn, addr = ssock.accept()" line), this can lead to weak encryption algorithms that are susceptible to brute force attacks especially with today's computation power.